

REPUBLIC OF RWANDA

EASTERN PROVINCE

KIREHE DISTRICT

KIGINA SECTOR

KIREHE ADVENTIST TECHNICAL SECONDARY SCHOOL

Quality Assurance Questions and Answers of Ending Units Assessments

From The Trainee Manual

Module Title & Code:

GENQA501 – Quality Assurance Application

Learning Outcomes:

1. **Perform Requirements Analysis**
2. **Test the System**
3. **Generate Test Documentation**

☐ **Learning Outcome 1: Perform Requirements Analysis**

You will learn:

- What QA is and its processes
- How to analyze a Terms of Reference (ToR)
- How to examine a requirement specification
- How to analyze an inception report

Includes:

- Functional vs Non-functional requirements
- Techniques like static testing, regression testing, performance testing
- **QA** vs **QC** differences

☐ **Learning Outcome 2: Test the System**

You will learn:

- How to prepare a test plan and testing environment
- How to perform various types of software testing
- Includes real-world testing tools and automation strategies

Learning Outcome 3: Generate Test Documentation

You will learn:

- How to consolidate test results
- How to prepare a **User Acceptance Testing (UAT)** report
- How to write recommendation reports

Assessment Types

- **Theoretical assessments:** Multiple-choice, True/False, short answer
- **Practical tasks:** Analyze ToR, review requirement specs, generate test reports

Tools & Methods Covered

- **Testing Tools:** JIRA, Selenium, JMeter, Git, Jenkins, OWASP ZAP, etc.
- **Techniques:** Exploratory testing, regression testing, automation, etc.
- **Standards:** ISO 9001, Six Sigma, IEEE, etc.

Quality Assurance (QA): Ibisobanuro birambuye mu Kinyarwanda n'Icyongereza

1. Definition of QA – Ibisobanuro bya Quality Assurance

Quality Assurance (QA) : is a **systematic process** used to make sure that a software product or service **meets the required quality standards**.

QA ni **uburyo buhoraho kandi bunoze** bukoreshwa mu gucunga no gukurikirana ko porogaramu cyangwa serivisi yakozwe **ihuye n'ubuziranenge bwifuzwa**.

Purpose:

- Kugenzura niba ibyo abakiriya bakeneye byubahirijwe.
- Gukumira amakosa (prevention), aho kuyakosora amaze kuba (detection).

2. Main Stages of QA – Intambwe z'ingenzi za QA

Ibi ni ibigize urugendo rwa QA nk'uko byasobanuwe:

1. **Analyze Requirements** – Gusuzuma ibisabwa
2. **Test Planning** – Gushyiraho gahunda y’igerageza
3. **Test Design** – Gutegura test cases
4. **Test Execution & Defect Reporting** – Gukora igerageza no gutanga raporo z’amakosa
5. **Re-tests & Regression Testing** – Gukora re-test na regression test
6. **Release Tests** – Gukora igenzura ry’ibisohoka

□ 3. Types of QA – Ubwoko bwa QA

1. **Functional QA:** Testing behavior of system
 - 🛠️ Kugenzura niba functionalities zikora neza (login, register, etc.)
2. **Non-Functional QA:** Performance, scalability, usability...
 - □ Nko kureba uko application yitwara ku bantu benshi (load testing)
3. **Automated QA:**
 - □ Koresha tools nka Selenium, JUnit gukora test zihoraho
4. **Manual QA:**
 - □ Umuntu akora test n’intoki
5. **Regression QA:**
 - ↻ Kugenzura niba ibindi bice bidahindutse nyuma yo gukosora
6. **UAT – User Acceptance Testing:**
 - 👤 Abakoresha bemeza niba software ibanyuze
7. **Compliance QA:**
 - ✓ Kugenzura niba application yubahiriza amategeko n’amabwiriza (GDPR, ISO)

□ 4. Standards za QA – Ibyiciro by’ubuziranenge bikoreshwa

1. **ISO 9001** – General Quality Management
2. **ISO/IEC 17025** – Laboratory QA
3. **Six Sigma** – Quality optimization
4. **IEEE Standards** – Software engineering principles

🛠️ Aya ni amahame agenga imikorere y’ubuziranenge mu bice byose bya software.

✂ 5. QA Methods – Uburyo bukoreshwa muri QA

1. **Audit** – Isuzuma ry’imigendekere y’imirimo
2. **Review** – Guhuza abakozi n’abakiriya ngo

the **Quality Assurance Application – GENQA501** manual:

✓ i. What is a Recommendation Report?

Recommendation Report ni inyandiko y’ibitekerezo n’ibisubizo byatanze hashingiwe ku isesengura ryakozwe nyuma yo gukora software testing.

It's a **formal document** that summarizes findings from the testing phase and provides **suggested actions** to improve or make decisions about a software system.

🔍 Ibarimo:

- Ibyabonetse mu bizamini (**findings**)
- Ibibazo byagaragaye (**Issues found**)
- Inama (**recommendations**) ku buryo software yakosorwa cyangwa yanozwa
- Ingaruka zo kutagira icyo gihinduka (**Impacts**)

✓ ii. What are the components of a Recommendation Report?

☐ **Ibigize recommendation report** (according to the manual) ni ibi bikurikira:

1. **Title Page** – Uru rupapuro rurimo:
 - Izina ry'inyandiko
 - Abo yateguwe na bo
 - Itariki
2. **Table of Contents** – Imbonerahamwe y'ibikubiye mu nyandiko
3. **Executive Summary** – Incamake y'ibyangenzi
 - Findings, key risks, inama z'ingenzi
4. **Introduction** – Intangiriro
 - Icyatumye report ikorwa, intego zayo
5. **Methodology** – Uburyo hakozwe assessment
 - Test plans, testing techniques, etc.
6. **Findings Summary** – Ibyabonetse mu igenzura
 - Defects, inconsistencies, bugs...
7. **Recommendations** – Inama zifatika
 - Icyo wakora ku kibazo runaka
8. **Justification** – Impamvu z'ayo mahitamo
 - Ni gute iyi recommendation izafasha
9. **Benefits and Impacts** – Inyungu n'ingaruka
 - Zizaturuka ku gushyira mu bikorwa recommendations
10. **Conclusion** – Umwanzuro
11. **Appendices** – Inyongera (test cases, charts, logs...)

✓ iii. What are the Benefits and Impacts of a Recommendation Report?

🎁 Inyungu (Benefits):

- ☐ It helps **decision-makers** take informed actions.

-  It ensures that **issues found during QA** are addressed.
-  Igabanya ibyago byo gukora software idahuye n'ibyifuzo.
-  It leads to **continuous improvement** of the system.
- ☐ Irakoreshwa nk'inyandiko ishingirwaho mu cyemezo cya final approval.

Impacts (Ingaruka):

-  Iyo utayikoze, project irashobora kugenda nabi cyangwa idasohoka.
-  Is saves time and money by preventing **recurring defects**.
-  Igira uruhare mu gutuma system iba reliable kandi efficient.

iv. Describe the Archive Test Artifacts

Test artifacts ni inyandiko cyangwa ibimenyetso biba byarakozwe igihe QA yakozwe (during software testing lifecycle).

 **Archive** means to **store these test artifacts** in an organized, retrievable manner for future use.

Examples of test artifacts to be archived:

1.  **Test Plans** – Document describing scope, approach, resources, and schedule of testing
2. ☐ **Test Cases** – Instructions to test a particular functionality
3. ☐ **Test Reports** – Summary of results from test executions
4.  **Defect Logs** – List of bugs found during testing
5.  **Metrics & KPIs** – Testing statistics (like pass/fail %, bug density)
6.  **User Acceptance Testing (UAT) Results**
7.  **Communication logs** – Emails, feedback, clarifications

Why archive test artifacts?

- For **compliance** and **auditing**
- Kugira ibimenyetso byo gusubiraho testing process
- Gufasha QA teams mu bindi bihe (future releases)
- Guhuza amakuru hagati y'abakozi bashya n'abasanze

Waba ushaka ko tugukorera **sample Recommendation Report** cyangwa **Test Artifacts archive template**? Nditeguye kugufasha.

No	Statement	Answer	Explanation
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1	QA is solely responsible for finding defects in software.	FALSE	QA prevents defects, not only finds them like QC.
2	QA should be performed only after development is complete.	FALSE	QA starts early (during planning & requirement analysis).
3	A high defect rate indicates poor software quality.	TRUE	Yes, many bugs = low quality.
4	QA teams are responsible for writing requirements and specifications.	FALSE	BA (Business Analyst) usually writes specs; QA just analyzes them.
5	QA teams should always strive for zero defects.	TRUE	That's a QA goal minimize defects as much as possible.
6	QA is only concerned with functional correctness.	FALSE	QA also includes performance, security, usability...
7	Automated testing tools can completely replace manual testing.	FALSE	Some tests still need human intuition (e.g., usability).
8	QA teams should be isolated from development teams to avoid conflicts of interest.	FALSE	They should collaborate with developers.
9	QA is primarily focused on preventing defects.	TRUE	Ni igikorwa cy'ubwirinzi (prevention) .
10	QA stages include planning, assurance, and control.	TRUE	These are part of QA lifecycle.
11	QC is proactive, QA is reactive.	FALSE	QA is proactive , QC is reactive.
12	ISO 9001 is a widely recognized QA standard.	TRUE	Yes, used worldwide for quality systems .
13	One of the main benefits of QA is customer satisfaction.	TRUE	QA ensures software meets expectations .
14	Statistical Process Control (SPC) is a tool used in QC only.	FALSE	SPC is used in both QC and QA.
15	QA uses methods like audits and process evaluations.	TRUE	Yes, QA uses reviews, audits, inspections .

16	Lean methodology is an example of QA standard.	FALSE	Lean is a general methodology , not specific to QA.
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LU1 Assessment: Perform Requirements Analysis

◆ Q1: True or False (Sobanukirwa buri gisubizo)

◆ Q2: Multiple Choice Questions (Circle the correct letter)

✓ Each answer is followed by a brief explanation.

i. Which of the following is NOT an example of industry-specific QA?

- a) Software QA
- b) Healthcare QA
- c) Financial QA
- d) **Retail QA** ✓

Explanation: Retail doesn't have a specific QA branch like **finance or health**.

ii. Which of the following is a benefit of manual testing?

- a) Faster test execution
- b) Higher test coverage
- c) **Ability to detect subtle defects** ✓
- d) Lower cost

Manual testing is better at detecting **human-related issues** (e.g., UI, UX).

iii. The main objective of quality assurance is:

- a) Proof of fitness product
- b) Inspection of quality of product
- c) Quality conformance
- d) **Customer satisfaction** ✓

QA aims at ensuring the **end user is satisfied**.

iv. The primary goal of analyzing an inception report is to:

- a) Assess the project's feasibility and risks
- b) Review the project's documentation

- c) Identify stakeholders
- d) **All of the above** ✓

Inception report analysis includes **all those activities**.

v. Purpose of documenting assumptions and constraints in inception report:

- a) To identify potential risks
- b) To clarify project scope
- c) To ensure project feasibility
- d) **All of the above** ✓

Zose zifatanye mu gutuma project igenda neza.

vi. Findings from inception report analysis are used to:

- a) Develop a project plan
- b) Identify potential risks
- c) Communicate project goals to stakeholders
- d) **All of the above** ✓

◆ **Q3: What is Quality Assurance (QA) in software development?**

Answer:

QA is a **systematic and proactive process** that ensures the software product meets **specified requirements and quality standards**. It focuses on **preventing defects** during development using **reviews, audits, and test strategies**.

◆ **Q4: What is the main difference between QA and QC?**

Aspect	QA	QC
Focus	Prevention of defects	Detection of defects
Timing	Happens during development	Happens after development
Type	Process-oriented	Product-oriented

◆ **Q5: Fill in the blank**

1. is part of quality management focused on providing confidence that quality requirements will be fulfilled.

Answer: c) Quality Assurance

2. The first step in analyzing an inception report is to perform a

Answer: a) Document review

◆ Q6: What is the difference between functional and non-functional requirements?

Functional Requirements	Non-Functional Requirements
Describe what the system should do	Describe how the system behaves
Based on user needs	Based on performance needs
Ex: login, data saving	Ex: response time, scalability

📁 Practical Assessment Task (Brief Summary)

You are working on a CRM system. You need to:

1. Implement QA processes
2. Analyze the **ToR document**
3. Review **requirement specifications**
4. Assess the **inception report**

Ibi byose bigufasha guhuza software n'ibyo abakiriya bakeneye.

✓ Learning Outcome 2: Test the System – End Assessment

◆ Q1: True or False

◆ Q2: Multiple Choice (Circle the correct letter)

#	Statement	Answer	Explanation
1	Test Plan defines what to test and how to test it.	True ✓	Test Plan ni igikoresho cy'ibanze mu QA gitanga uburyo bwo gukora tests.
2	Test environment includes hardware, software, and network setup.	True ✓	Birimo ibikoresho byose bikoreshwa mu igerageza.
3	Regression testing is used to check new features.	False ✗	Regression testing ireba niba ibisanzwe bitangiritse nyuma y'impinduka.
4	Load testing checks how the system performs under extreme conditions.	False ✗	Iyo ni stress testing ; load testing ireba uko system yitwara mu bikenewe bisanzwe .
5	Automation testing eliminates the need for manual testing.	False ✗	Nubwo automation ifasha, hari tests zigomba gukorwa n'abantu (e.g., usability tests).
6	Test cases are written before the actual testing starts.	True ✓	Test cases zitegurwa mbere y'igihe, zerekana ibyo tuzapima n'uko.
7	Functional testing evaluates system performance under load.	False ✗	Performance testing niyo igenzura uko system yitwara ku load.
8	Smoke testing is done to check critical features before deeper testing.	True ✓	Ni "sanity check" ireba niba ibintu by'ibanze bikora mbere yo kujya kure.
9	Integration testing is done after unit testing.	True ✓	Kuko ireba uko modules zifatanya, nyuma yo gupima buri imwe.
10	Defect reporting includes status, severity, and description.	True ✓	Iyi raporo irimo ibintu byose by'ingirakamaro kugira ngo bug ikosorwe.

i. What is the main objective of a test plan?

- a) Estimate the cost of testing
- b) Describe the testing objectives and methods
- c) Assign tasks to developers
- d) Install software

→ **Correct answer: b) Describe the testing objectives and methods**

ii. Which testing type checks if new changes broke existing features?

- a) Unit Testing
- b) Smoke Testing

- c) **Regression Testing** ✓
d) Integration Testing

iii. Which one is a non-functional testing type?

- a) System Testing
b) Unit Testing
c) Performance Testing ✓
d) UAT

iv. Which of the following is NOT a test environment component?

- a) Server
b) Database
c) Programmer
d) Operating System

→ **Correct answer: c) Programmer**

A programmer is **not** part of the test environment setup.

v. What is the purpose of smoke testing?

- a) To test database integrity
b) To verify core functionalities work
c) To test all modules deeply
d) To do performance testing

→ **Correct answer: b) To verify core functionalities work**

◆ **Q3: Short Answer**

i. What is a test case?

A **test case** is a set of actions and expected results designed to verify whether a specific functionality works as intended.

Urugero: Kugenzura niba **login** ikora neza uko bikwiye.

Includes:

- Test case name
- Steps to execute
- Input data
- Expected result
- Actual result
- Pass/Fail status

ii. What is the difference between manual and automated testing?

Manual Testing	Automated Testing
Performed by human testers	Uses scripts and tools
Better for exploratory and usability testing	Better for repetitive and regression tests
Slower and prone to human error	Faster and more consistent

iii. Define the test environment.

A **test environment** is the **hardware and software setup** where testing is done.

It includes: servers, operating systems, browsers, databases, networks — byose bikenerwa ngo testing ibashe gukorwa neza.

iv. What is defect reporting in QA?

It is the process of **recording and tracking software bugs** found during testing.

Report usually includes:

- Defect ID
- Title
- Severity
- Steps to reproduce
- Assigned developer
- Status (Open, In Progress, Closed)

◆ Q4: Fill in the Blank (Choose the correct answer)

1. A _____ is used to check whether the system works after a recent code change.

✓ Answer: Regression Test

2. The testing done by end-users to accept the software is called _____.

✓ Answer: User Acceptance Testing (UAT)

3. _____ testing ensures the system performs well under pressure.

✓ Answer: Stress

4. In testing, _____ means retesting a fixed bug.

✓ Answer: Re-testing

5. _____ testing is used to verify that the system performs under expected user load.

✓ Answer: Load

✓ Practical Task Overview (from manual)

Scenario: You are testing a software system for a local government.

You are required to:

1. Prepare a **test plan**
2. Set up a **test environment**
3. Perform **manual or automated testing**
4. Write a **defect report**
5. Present your **test results**

Ibi byose bigamije kureba niba software yujuje ibisabwa n'abakoresha.

✓ Learning Outcome 3: Generate Test Documentation – End Assessment

◆ Q1: True or False (Sobanukirwa buri gisubizo)

1. True Or False

#	Statement	Answer	Explanation
1	Test documentation includes test plans, test cases, and test reports.	✓ True	All are QA test artifacts — byemeza uburyo tests zakozwe n'uko zasohotse.
2	Consolidated test results summarize all testing activities.	✓ True	Yes, they summarize test execution outcomes (passed/failed/blocked).
3	UAT report is written by developers.	✗ False	It's written by QA teams or users after acceptance testing.
4	Recommendation report includes suggestions after testing.	✓ True	Yes, it proposes actions to improve software after tests.
5	Only technical people should read test documentation.	✗ False	Also stakeholders, clients, and management need these reports.
6	Archiving test artifacts is unnecessary after deployment.	✗ False	Archiving is important for future reference, audits, and traceability.

7	A test case should have input data and expected output.	✓ True	Yes, it shows what to test, how, and expected results.
8	A well-prepared recommendation report helps improve decision-making.	✓ True	Yes, it gives facts and advice based on test findings.
9	UAT focuses on performance testing.	✗ False	UAT focuses on user needs and satisfaction , not system performance.
10	Test summary report includes defect details.	✓ True	It includes number of defects, severity, resolution status, etc.

◆ Q2: Multiple Choice Questions (Circle the correct letter)

i. Which document summarizes all testing results and metrics?

- a) Test Case
- b) UAT Report
- c) **Test Summary Report** ✓
- d) Requirement Specification

Iri tangazo ririmo **pass/fail** %, **defects**, **test coverage**, etc.

ii. What is the purpose of the UAT report?

- a) To test internal code
- b) To finalize the UI
- c) **To validate the system meets user needs** ✓
- d) To list bugs

iii. What is NOT included in a recommendation report?

- a) Findings
- b) Recommendations
- c) System Design
- d) **Hardware Purchase Orders** ✓

Ibi ntibirebana na QA documentation.

iv. Which of the following is a reason to archive test artifacts?

- a) Free up RAM
- b) Save money
- c) **Enable traceability and audit** ✓
- d) Avoid testing again

v. What is a key feature of test documentation?

- a) Unstructured notes

- b) Verbal feedback
- c) **Traceability and clarity** ✓
- d) Random screenshots

◆ Q3: Short Answer Questions (Explain clearly)

i. What is a Test Summary Report?

It is a **document that summarizes the results of all tests performed**, including:

- Total test cases executed
- Passed/Failed
- Defects found
- Testing coverage
- Final conclusion (ready for release or not)

Tip: *It is presented to stakeholders before making release decisions.*

ii. What is a User Acceptance Testing (UAT) Report?

UAT report is a **document written after end-users test the system** to confirm that:

- The system works as expected
- User needs have been met
- Major defects have been fixed

🔗 *It gives a “go” or “no-go” decision.*

iii. Why do we write a Recommendation Report in QA?

We write a recommendation report to:

- **Summarize findings** from testing
- Give **suggestions** for improvement
- Help **stakeholders** make informed decisions
- Propose **next actions** (e.g., fix bugs, deploy, or do more testing)

🔗 *It is based on test results and helps improve project quality.*

iv. What is meant by “archiving test artifacts”?

To **archive test artifacts** means to **store and organize** all testing documents for:

- Future reference
- Compliance checks
- Traceability
- Knowledge transfer

Examples include: test cases, test plans, defect reports, summary reports, UAT docs.

◆ Q4: Fill in the Blanks (Choose the correct words)

1. A _____ report contains the result of tests performed by users.

✓ Answer: UAT

2. A _____ summarizes testing efforts and outcomes.

✓ Answer: Test Summary Report

3. A _____ includes suggestions after system testing.

✓ Answer: Recommendation Report

4. A complete test case should have _____ and _____.

✓ Answer: input data, expected result

5. Test artifacts should be stored in _____ for future access.

✓ Answer: archive

◆ Practical Task Summary from LU3 Manual

Scenario: After finishing system testing, you're asked to:

1. ✦ Consolidate test results
2. □ Prepare a **UAT report**
3. 📄 Write a **recommendation report**
4. 🗄️ Archive all test artifacts

🔗 All these documents help in making informed decisions about releasing or improving the software.